Homework 5

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Overview

In this homework assignment, you will explore, analyze and model a data set containing information on approximately 12,000 commercially available wines. The variables are mostly related to the chemical properties of the wine being sold. The response variable is the number of sample cases of wine that were purchased by wine distribution companies after sampling a wine. These cases would be used to provide tasting samples to restaurants and wine stores around the United States. The more sample cases purchased, the more likely is a wine to be sold at a high end restaurant. A large wine manufacturer is studying the data in order to predict the number of wine cases ordered based upon the wine characteristics. If the wine manufacturer can predict the number of cases, then that manufacturer will be able to adjust their wine offering to maximize sales.

1. DATA EXPLORATION

Explore, analyze and model a data set containing approximately 8000 records representing a customer at an auto insurance company. Each record has two response variables. The first response variable, TARGET_FLAG, is a 1 or a 0. A "1" means that the person was in a car crash. A zero means that the person was not in a car crash. The second response variable is TARGET_AMT. This value is zero if the person did not crash their car. But if they did crash their car, this number will be a value greater than zero.

VARIABLE NAME DEFINITION THEORETICAL EFFECT

- INDEX: Identification Variable (do not use) None
- TARGET: Number of Cases Purchased None
- AcidIndex: Proprietary method of testing total acidity of wine by using a weighted average
- Alcohol: Alcohol Content
- Chlorides: Chloride content of wine
- CitricAcid: Citric Acid Content
- Density: Density of Wine
- FixedAcidity: Fixed Acidity of Wine
- FreeSulfurDioxide: Sulfur Dioxide content of wine
- LabelAppeal: Marketing Score indicating the appeal of label design for consumers. High numbers suggest customers like the label design. Negative numbers suggest customes don't like the design. (Many consumers purchase based on the visual appeal of the wine label design. Higher numbers suggest better sales.)
- ResidualSugar: Residual Sugar of wine
- STARS: Wine rating by a team of experts. 4 Stars = Excellent, 1 Star = Poor (A high number of stars suggests high sales)
- Sulphates: Sulfate conten of wine
- TotalSulfurDioxide: Total Sulfur Dioxide of Wine
- VolatileAcidity: Volatile Acid content of wine
- pH: pH of wine

Dataset The dataset contains 16 columns and 12,795 rows. An initial review shows a number of missing values, with 8 columns containing NAs. We also observe that several numerical features include negative values and no columns with near zero variance.

```
train<- read.csv("https://raw.githubusercontent.com/Chung-Brandon/621-Data-Mining/refs/heads/main/wine-

test <- read.csv("https://raw.githubusercontent.com/Chung-Brandon/621-Data-Mining/refs/heads/main/wine-
```

```
glimpse(train)
```

```
## Rows: 12,795
## Columns: 16
## $ INDEX              <int> 1, 2, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 16, 17, 19~
## $ TARGET             <int> 3, 3, 5, 3, 4, 0, 0, 4, 3, 6, 0, 4, 3, 7, 4, 0, 0, ~
## $ FixedAcidity       <dbl> 3.2, 4.5, 7.1, 5.7, 8.0, 11.3, 7.7, 6.5, 14.8, 5.5,~
## $ VolatileAcidity    <dbl> 1.160, 0.160, 2.640, 0.385, 0.330, 0.320, 0.290, -1~
## $ CitricAcid         <dbl> -0.98, -0.81, -0.88, 0.04, -1.26, 0.59, -0.40, 0.34~
## $ ResidualSugar      <dbl> 54.20, 26.10, 14.80, 18.80, 9.40, 2.20, 21.50, 1.40~
## $ Chlorides          <dbl> -0.567, -0.425, 0.037, -0.425, NA, 0.556, 0.060, 0.~
## $ FreeSulfurDioxide  <dbl> NA, 15, 214, 22, -167, -37, 287, 523, -213, 62, 551~
## $ TotalSulfurDioxide <dbl> 268, -327, 142, 115, 108, 15, 156, 551, NA, 180, 65~
## $ Density            <dbl> 0.99280, 1.02792, 0.99518, 0.99640, 0.99457, 0.9994~
## $ pH                 <dbl> 3.33, 3.38, 3.12, 2.24, 3.12, 3.20, 3.49, 3.20, 4.9~
## $ Sulphates          <dbl> -0.59, 0.70, 0.48, 1.83, 1.77, 1.29, 1.21, NA, 0.26~
## $ Alcohol            <dbl> 9.9, NA, 22.0, 6.2, 13.7, 15.4, 10.3, 11.6, 15.0, 1~
## $ LabelAppeal        <int> 0, -1, -1, -1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 2, 0, 0, ~
## $ AcidIndex          <int> 8, 7, 8, 6, 9, 11, 8, 7, 6, 8, 5, 10, 7, 8, 9, 8, 9~
## $ STARS              <int> 2, 3, 3, 1, 2, NA, NA, 3, NA, 4, 1, 2, 2, 3, NA, NA~
```

```
summary(train)
```

```
##      INDEX           TARGET       FixedAcidity     VolatileAcidity
##  Min.   :    1   Min.   :0.000   Min.   :-18.100   Min.   :-2.7900
##  1st Qu.: 4038   1st Qu.:2.000   1st Qu.:  5.200   1st Qu.: 0.1300
##  Median : 8110   Median :3.000   Median :  6.900   Median : 0.2800
##  Mean   : 8070   Mean   :3.029   Mean   :  7.076   Mean   : 0.3241
##  3rd Qu.:12106   3rd Qu.:4.000   3rd Qu.:  9.500   3rd Qu.: 0.6400
##  Max.   :16129   Max.   :8.000   Max.   : 34.400   Max.   : 3.6800
##
##    CitricAcid      ResidualSugar        Chlorides       FreeSulfurDioxide
##  Min.   :-3.2400   Min.   :-127.800   Min.   :-1.1710   Min.   :-555.00
##  1st Qu.: 0.0300   1st Qu.:  -2.000   1st Qu.:-0.0310   1st Qu.:   0.00
##  Median : 0.3100   Median :   3.900   Median : 0.0460   Median :  30.00
##  Mean   : 0.3084   Mean   :   5.419   Mean   : 0.0548   Mean   :  30.85
##  3rd Qu.: 0.5800   3rd Qu.:  15.900   3rd Qu.: 0.1530   3rd Qu.:  70.00
##  Max.   : 3.8600   Max.   : 141.150   Max.   : 1.3510   Max.   : 623.00
##                    NA's   :616        NA's   :638       NA's   :647
##  TotalSulfurDioxide    Density             pH          Sulphates
##  Min.   :-823.0     Min.   :0.8881   Min.   :0.480   Min.   :-3.1300
##  1st Qu.:  27.0     1st Qu.:0.9877   1st Qu.:2.960   1st Qu.: 0.2800
##  Median : 123.0     Median :0.9945   Median :3.200   Median : 0.5000
##  Mean   : 120.7     Mean   :0.9942   Mean   :3.208   Mean   : 0.5271
##  3rd Qu.: 208.0     3rd Qu.:1.0005   3rd Qu.:3.470   3rd Qu.: 0.8600
##  Max.   :1057.0     Max.   :1.0992   Max.   :6.130   Max.   : 4.2400
##  NA's   :682                         NA's   :395     NA's   :1210
##     Alcohol       LabelAppeal          AcidIndex          STARS
```

```
##  Min.   :-4.70   Min.   :-2.000000   Min.   : 4.000   Min.   :1.000
##  1st Qu.: 9.00   1st Qu.:-1.000000   1st Qu.: 7.000   1st Qu.:1.000
##  Median :10.40   Median : 0.000000   Median : 8.000   Median :2.000
##  Mean   :10.49   Mean   :-0.009066   Mean   : 7.773   Mean   :2.042
##  3rd Qu.:12.40   3rd Qu.: 1.000000   3rd Qu.: 8.000   3rd Qu.:3.000
##  Max.   :26.50   Max.   : 2.000000   Max.   :17.000   Max.   :4.000
##  NA's   :653                                          NA's   :3359
```

```
colSums(is.na(train))
```

```
##              INDEX             TARGET       FixedAcidity    VolatileAcidity
##                  0                  0                  0                  0
##         CitricAcid      ResidualSugar          Chlorides  FreeSulfurDioxide
##                  0                616                638                647
## TotalSulfurDioxide            Density                 pH          Sulphates
##                682                  0                395               1210
##            Alcohol        LabelAppeal          AcidIndex              STARS
##                653                  0                  0               3359
```

```
nzv <- nearZeroVar(train, saveMetrics = TRUE)
nzv$nzv
```

```
##  [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [13] FALSE FALSE FALSE
```

Distributions

- We see that most variables have normal distribution, like Alcohol, FixedAcidity, Density, pH, Sulphates, VolatileAcidity have strong central peaks.
- FreeSulfurDioxide and TotalSulfurDioxide extend from negative to positive values. This indicates data quality issues or special cases that may require transformation.
- Variable AcidIndex, STARS, and LabelAppeal shows a multimodal distribution.

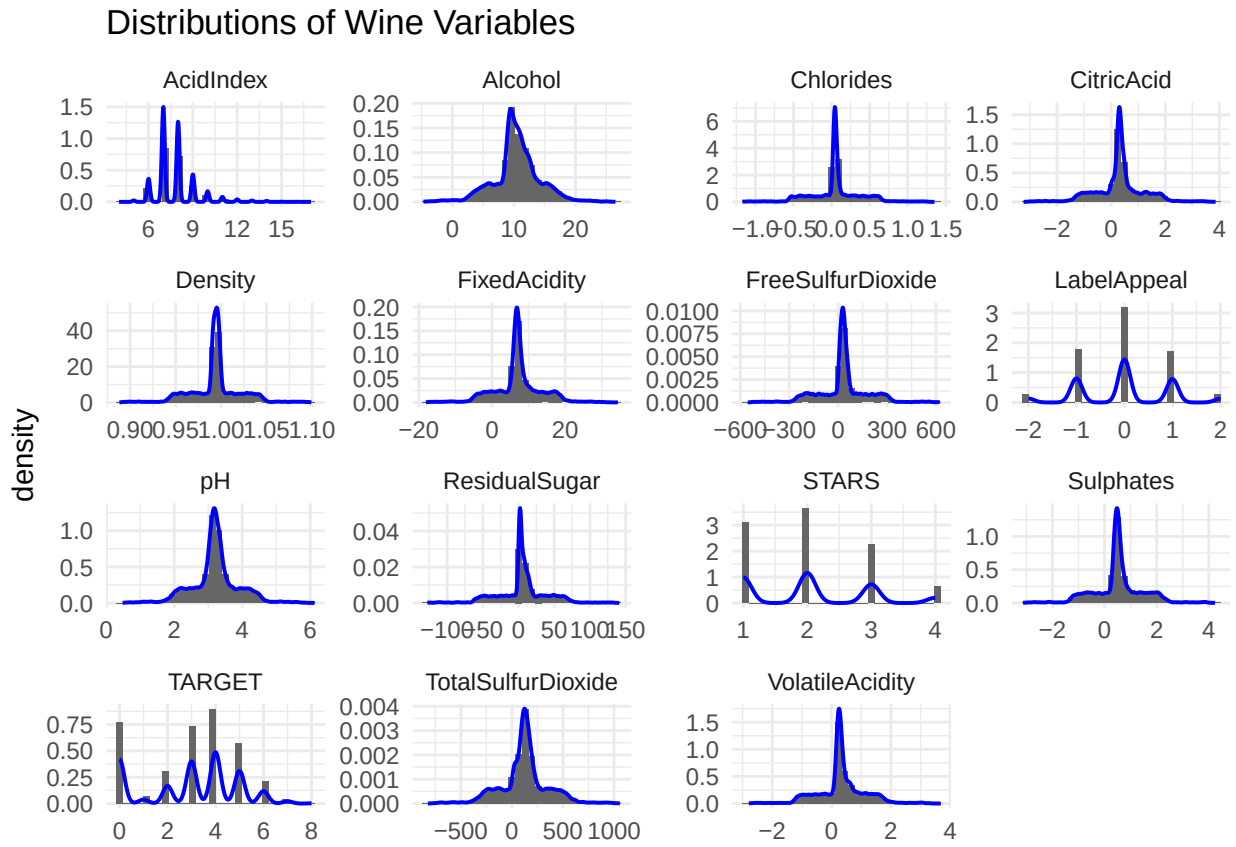
```
train_long <- train %>%
  pivot_longer(
    cols = everything(),
    names_to  = "variable",
    values_to = "value"
  )

ggplot(train_long, aes(x = value)) +
  geom_histogram(aes(y = ..density..),
                 bins = 30,
                 fill = "grey40") +
  geom_density(color = "blue", linewidth = 0.7, na.rm = TRUE) +
  facet_wrap(~ variable, scales = "free", ncol = 4) +
  labs(x = NULL, y = "density", title = "Distributions of Wine Variables") +
  theme_minimal()
```

```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## Warning: Removed 8200 rows containing non-finite outside the scale range
```

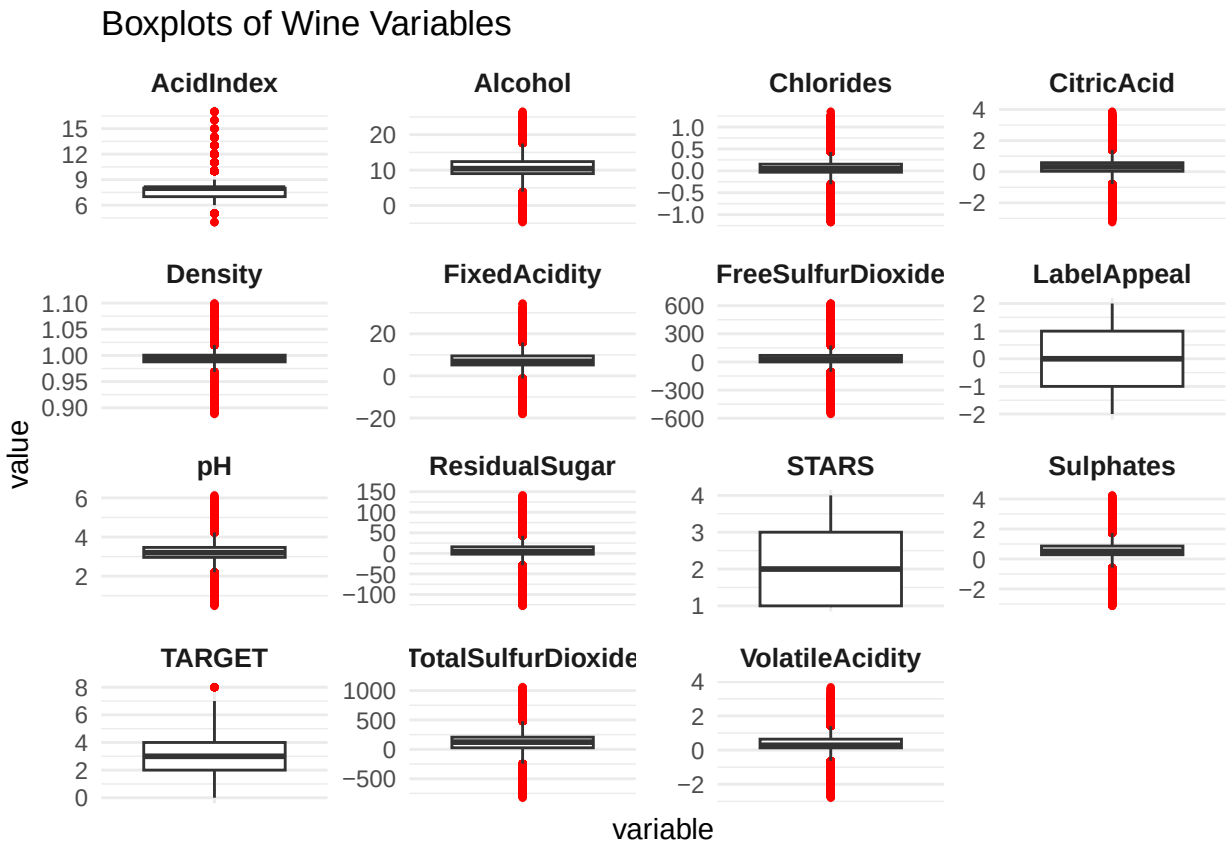
```
## (`stat_bin()`).
```



Boxplots

- Several variables have extreme outliers.
- ResidualSugar, TotalSulfurDioxide, FreeSulfurDioxide have very large positive and negative outliers.

```
# Boxplots
ggplot(train_long, aes(x = variable, y = value)) +
  geom_boxplot(outlier.color = "red", outlier.size = 0.8, na.rm = TRUE) +
  facet_wrap(~ variable, scales = "free", ncol = 4) +
  labs(
    title = "Boxplots of Wine Variables"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_blank(),
    strip.text = element_text(size = 10, face = "bold")
  )
```

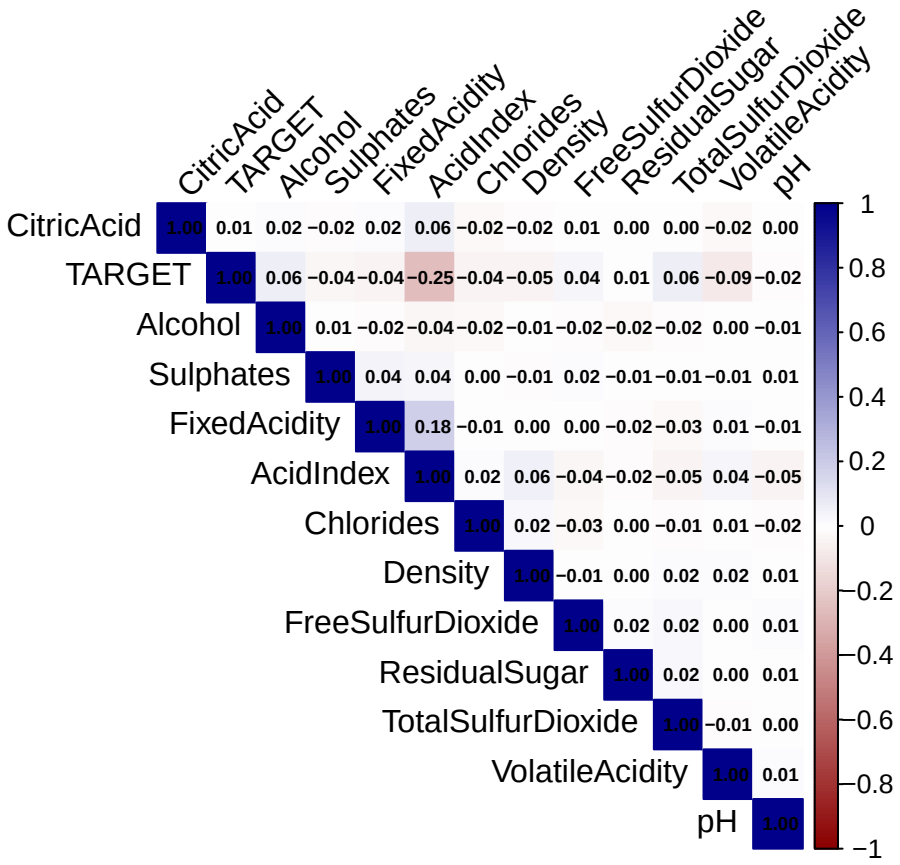


Correlation matrix From our correlation matrix we can see that Acidindex seems to have the greatest impact on our TARGET, the number of cases purchased, this finding will warrent further investigation. Also, we can note here that other numeric variables do not seem to have large impacts on the TARGET variable at first glance.

```
# Correlation matrix for the numeric variables, excluding STARS and LabelAppeal as we are treating them

corr_matrix <- cor(train[, c("TARGET","FixedAcidity", "VolatileAcidity", "CitricAcid","ResidualSugar","

corrplot(
  corr_matrix,
  method = "color",
  type = "upper",
  order = "hclust",
  addCoef.col = "black",
  tl.col = "black",
  tl.srt = 45,
  number.cex = 0.6,
  col = colorRampPalette(c("darkred", "white", "darkblue"))(200)
)
```



2. DATA Preparation

Missing Values For the missing values in the STARS variable, this implicates that the wine is unranked. To capture the unranked status we have imputated NA values in STARS as "unranked". MICE imputation here would imply rankings for the wine where rankings may not be appropriate.

```
train <- train %>%
  mutate(STARS = ifelse(is.na(STARS), "unranked", STARS),
         STARS = as.factor(STARS))

train <- train %>%
  mutate(LabelAppeal = as.factor(LabelAppeal))
```

We use mice package to impute the remaining missing values in the dataset.

```
# Impute missing data using mice

set.seed(123)
imp <- mice(train, m = 5, method = "pmm")
```

```
##
##  iter imp variable
##   1   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   1   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   1   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   1   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   1   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
```

```
##   2   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   2   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   2   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   2   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   2   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   3   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   3   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   3   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   3   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   3   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   4   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   4   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   4   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   4   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   4   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   5   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   5   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   5   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   5   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##   5   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
```

```
train_mice <- complete(imp)
```

```
colSums(is.na(train_mice))
```

```
##             TARGET       FixedAcidity    VolatileAcidity         CitricAcid
##                  0                  0                  0                  0
##      ResidualSugar          Chlorides  FreeSulfurDioxide TotalSulfurDioxide
##                  0                  0                  0                  0
##            Density                 pH          Sulphates            Alcohol
##                  0                  0                  0                  0
##        LabelAppeal          AcidIndex              STARS
##                  0                  0                  0
```

Corrections were also applied to the test data

```
test <- test %>%
  mutate(STARS = ifelse(is.na(STARS), "unranked", STARS),
         STARS = as.factor(STARS))

test <- test %>%
  mutate(LabelAppeal = as.factor(LabelAppeal))
```

The testing data also had missing data used mice a well to impute missing data without imputing data to the target variables. The target variables were turned to empty strings to avoid mice from filling in those variables.

```
# specify methods for each column
methVar <- make.method(test)

# set target columns to empty strings "" so mice will not touch them
methVar[methVar != ""] <- "pmm"
methVar["TARGET"] <- ""

imp <- (mice(test, m = 5, method = methVar, seed = 123))
```

```
##
```

```
##  iter imp variable
##    1   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    1   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    1   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    1   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    1   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    2   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    2   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    2   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    2   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    2   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    3   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    3   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    3   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    3   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    3   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    4   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    4   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    4   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    4   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    4   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    5   1  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    5   2  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    5   3  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    5   4  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
##    5   5  ResidualSugar  Chlorides  FreeSulfurDioxide  TotalSulfurDioxide  pH  Sulphates  Alcohol
```

```
## Warning: Number of logged events: 1
```

```
test_imp <- complete(imp)


colSums(is.na(test_imp))
```

```
##             TARGET       FixedAcidity    VolatileAcidity         CitricAcid
##               3335                  0                  0                  0
##      ResidualSugar          Chlorides  FreeSulfurDioxide TotalSulfurDioxide
##                  0                  0                  0                  0
##            Density                 pH          Sulphates            Alcohol
##                  0                  0                  0                  0
##        LabelAppeal          AcidIndex              STARS
##                  0                  0                  0
```

Outliers Many numerical features have somewhat unreasonable negative values, we decided to shift the variables that had negative values to have the lowest values just above 0.

```
shift_to_positive <- function(x, eps = 0.01){
  x - min(x, na.rm = TRUE) + eps
}
```

```
chem_vars <- c(
  "FixedAcidity",
  "VolatileAcidity",
  "CitricAcid",
  "ResidualSugar",
  "Chlorides",
```

```
  "FreeSulfurDioxide",
  "TotalSulfurDioxide",
  "Sulphates",
  "Alcohol"
)
```

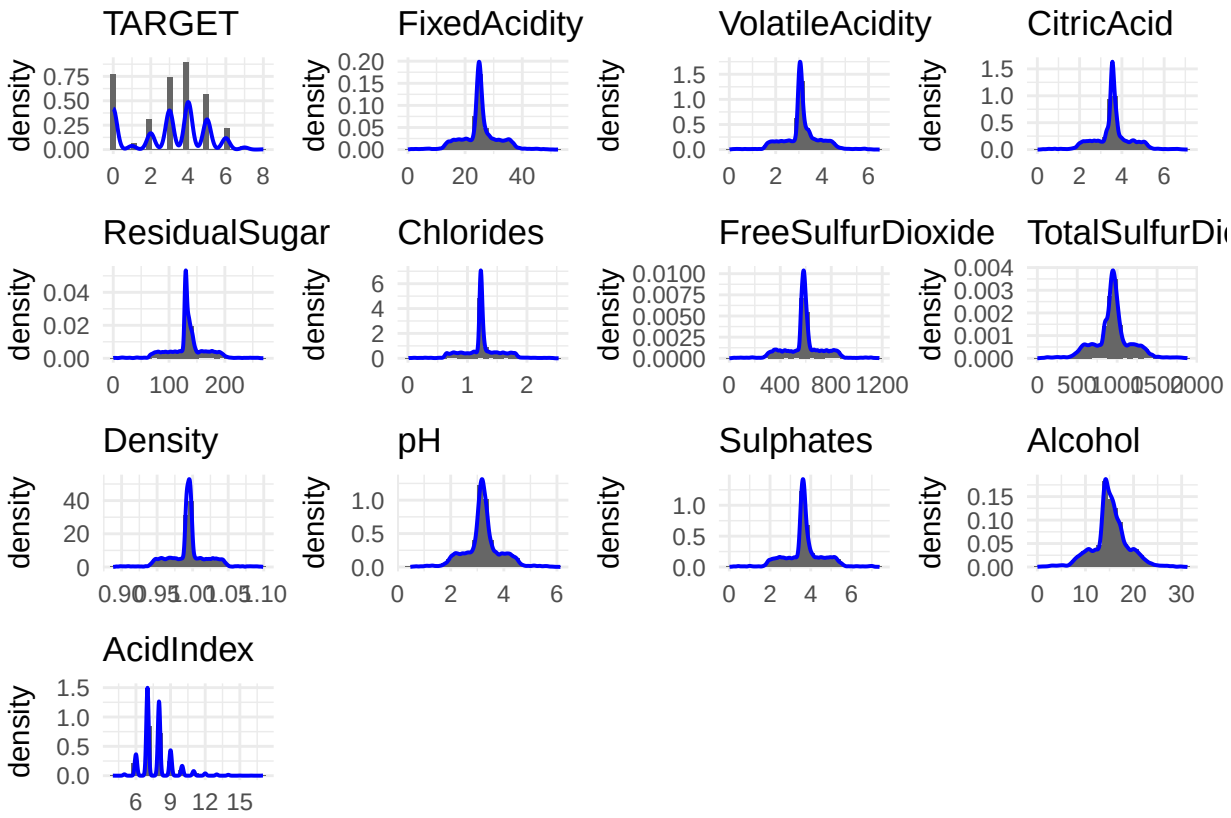
```
train_mice[chem_vars] <- lapply(train_mice[chem_vars], shift_to_positive)
test_imp[chem_vars]   <- lapply(test_imp[chem_vars], shift_to_positive)
```

Density plot to recheck distribution of numeric variables.

```
# Select numeric variables only (density requires numeric)
num_vars <- names(train_mice)[sapply(train_mice, is.numeric)]

plot_list <- lapply(num_vars, function(var) {
  ggplot(train_mice, aes(x = .data[[var]])) +
    geom_histogram(aes(y = ..density..),
                   bins = 30,
                   fill = "grey40") +
    geom_density(color = "blue", linewidth = 0.7, na.rm = TRUE) +
    labs(title = var, x = NULL, y = "density") +
    theme_minimal()
})

wrap_plots(plot_list, ncol = 4)
```



Models

Poisson Regression Model 1- baseline with all variables.

```
Model_p1<-glm(TARGET~., data=train_mice, family=poisson)
summary(Model_p1)
```

```
##
## Call:
## glm(formula = TARGET ~ ., family = poisson, data = train_mice)
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)          1.464e+00  2.057e-01   7.114 1.13e-12 ***
## FixedAcidity         3.663e-05  8.199e-04   0.045 0.964364
## VolatileAcidity     -3.039e-02  6.529e-03  -4.655 3.25e-06 ***
## CitricAcid           4.921e-03  5.899e-03   0.834 0.404145
## ResidualSugar        1.095e-04  1.506e-04   0.727 0.467338
## Chlorides           -3.718e-02  1.603e-02  -2.319 0.020398 *
## FreeSulfurDioxide    9.333e-05  3.422e-05   2.727 0.006387 **
## TotalSulfurDioxide   8.332e-05  2.204e-05   3.781 0.000156 ***
## Density             -2.598e-01  1.918e-01  -1.355 0.175437
## pH                  -1.198e-02  7.540e-03  -1.589 0.112148
## Sulphates           -1.373e-02  5.465e-03  -2.512 0.012018 *
## Alcohol              3.386e-03  1.374e-03   2.465 0.013698 *
## LabelAppeal-1        2.353e-01  3.799e-02   6.195 5.84e-10 ***
## LabelAppeal0         4.256e-01  3.705e-02  11.488  < 2e-16 ***
## LabelAppeal1         5.579e-01  3.769e-02  14.800  < 2e-16 ***
## LabelAppeal2         6.970e-01  4.244e-02  16.424  < 2e-16 ***
## AcidIndex           -7.960e-02  4.576e-03 -17.395  < 2e-16 ***
## STARS2               3.191e-01  1.436e-02  22.227  < 2e-16 ***
## STARS3               4.384e-01  1.562e-02  28.061  < 2e-16 ***
## STARS4               5.585e-01  2.167e-02  25.771  < 2e-16 ***
## STARSunranked       -7.665e-01  1.954e-02 -39.227  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 22861  on 12794  degrees of freedom
## Residual deviance: 13632  on 12774  degrees of freedom
## AIC: 45616
##
## Number of Fisher Scoring iterations: 6
```

Poisson Regression Model 2- with chemical variables.

```
set.seed(123)

Model_p2<- glm(TARGET ~ .-FixedAcidity -VolatileAcidity -CitricAcid -ResidualSugar -Chlorides -FreeSulfu
summary(Model_p2)
```

```
##
## Call:
## glm(formula = TARGET ~ . - FixedAcidity - VolatileAcidity - CitricAcid -
##     ResidualSugar - Chlorides - FreeSulfurDioxide - TotalSulfurDioxide -
```

```
##     Sulphates - Alcohol, family = poisson, data = train_mice)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.509851   0.197404   7.649 2.03e-14 ***
## Density       -0.261799   0.191629  -1.366   0.1719
## pH            -0.012510   0.007531  -1.661   0.0967 .
## LabelAppeal-1  0.236131   0.037978   6.218 5.05e-10 ***
## LabelAppeal0   0.425697   0.037042  11.492  < 2e-16 ***
## LabelAppeal1   0.558043   0.037674  14.812  < 2e-16 ***
## LabelAppeal2   0.696882   0.042422  16.427  < 2e-16 ***
## AcidIndex     -0.081662   0.004505 -18.128  < 2e-16 ***
## STARS2         0.321247   0.014349  22.389  < 2e-16 ***
## STARS3         0.443417   0.015588  28.445  < 2e-16 ***
## STARS4         0.564483   0.021611  26.121  < 2e-16 ***
## STARSunranked -0.770660   0.019530 -39.461  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 22861  on 12794  degrees of freedom
## Residual deviance: 13696  on 12783  degrees of freedom
## AIC: 45662
##
## Number of Fisher Scoring iterations: 6
```

Multiple linear regression Model- baseline with all variables

```
Model_M<-lm(TARGET~.,data = train_mice)
summary(Model_M)
```

```
##
## Call:
## lm(formula = TARGET ~ ., data = train_mice)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.9582 -0.8639  0.0212  0.8384  6.1713
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)         4.227e+00  4.633e-01   9.123  < 2e-16 ***
## FixedAcidity        6.107e-04  1.858e-03   0.329  0.74247
## VolatileAcidity    -9.459e-02  1.477e-02  -6.403 1.58e-10 ***
## CitricAcid          1.655e-02  1.343e-02   1.232  0.21805
## ResidualSugar       3.104e-04  3.421e-04   0.907  0.36426
## Chlorides          -1.176e-01  3.628e-02  -3.242  0.00119 **
## FreeSulfurDioxide   2.768e-04  7.786e-05   3.555  0.00038 ***
## TotalSulfurDioxide  2.392e-04  4.983e-05   4.801 1.60e-06 ***
## Density            -8.032e-01  4.357e-01  -1.844  0.06527 .
## pH                 -3.020e-02  1.703e-02  -1.773  0.07623 .
## Sulphates          -3.787e-02  1.238e-02  -3.058  0.00223 **
## Alcohol             1.151e-02  3.113e-03   3.696  0.00022 ***
## LabelAppeal-1       3.600e-01  6.283e-02   5.729 1.03e-08 ***
```

```
## LabelAppeal0        8.276e-01  6.127e-02  13.508  < 2e-16 ***
## LabelAppeal1        1.291e+00  6.400e-02  20.177  < 2e-16 ***
## LabelAppeal2        1.884e+00  8.432e-02  22.340  < 2e-16 ***
## AcidIndex          -1.994e-01  9.100e-03 -21.909  < 2e-16 ***
## STARS2              1.035e+00  3.259e-02  31.776  < 2e-16 ***
## STARS3              1.600e+00  3.766e-02  42.497  < 2e-16 ***
## STARS4              2.286e+00  5.975e-02  38.267  < 2e-16 ***
## STARSunranked      -1.364e+00  3.291e-02 -41.450  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.305 on 12774 degrees of freedom
## Multiple R-squared:  0.5415, Adjusted R-squared:  0.5408
## F-statistic: 754.2 on 20 and 12774 DF,  p-value: < 2.2e-16
```

Multiple Linear regression with chemical variables

```
set.seed(123)

Model_M2<- lm(TARGET ~ .-FixedAcidity -VolatileAcidity -CitricAcid -ResidualSugar -Chlorides -FreeSulfu
summary(Model_M2)
```

```
##
## Call:
## lm(formula = TARGET ~ . - FixedAcidity - VolatileAcidity - CitricAcid -
##     ResidualSugar - Chlorides - FreeSulfurDioxide - TotalSulfurDioxide -
##     Sulphates - Alcohol, data = train_mice)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.9293 -0.8687  0.0420  0.8399  6.2716
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.417615   0.445585   9.914  < 2e-16 ***
## Density       -0.835889   0.437236  -1.912   0.0559 .
## pH            -0.032589   0.017092  -1.907   0.0566 .
## LabelAppeal-1  0.358115   0.063074   5.678 1.39e-08 ***
## LabelAppeal0   0.824040   0.061504  13.398  < 2e-16 ***
## LabelAppeal1   1.286312   0.064227  20.028  < 2e-16 ***
## LabelAppeal2   1.876985   0.084641  22.176  < 2e-16 ***
## AcidIndex     -0.205824   0.008946 -23.006  < 2e-16 ***
## STARS2         1.041854   0.032703  31.858  < 2e-16 ***
## STARS3         1.615565   0.037753  42.792  < 2e-16 ***
## STARS4         2.306533   0.059892  38.511  < 2e-16 ***
## STARSunranked -1.376478   0.033011 -41.697  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.311 on 12783 degrees of freedom
## Multiple R-squared:  0.5374, Adjusted R-squared:  0.537
## F-statistic:  1350 on 11 and 12783 DF,  p-value: < 2.2e-16
```

Negative Binomial Model with all variables

```
Model_N<-glm.nb(TARGET~.,data=train_mice)
```

```
## Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
## control$trace > : iteration limit reached
## Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
## control$trace > : iteration limit reached
```

```
summary(Model_N)
```

```
##
## Call:
## glm.nb(formula = TARGET ~ ., data = train_mice, init.theta = 40803.08215,
##     link = log)
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)          1.464e+00  2.057e-01   7.114 1.13e-12 ***
## FixedAcidity         3.662e-05  8.199e-04   0.045 0.964371
## VolatileAcidity     -3.039e-02  6.529e-03  -4.655 3.25e-06 ***
## CitricAcid           4.921e-03  5.899e-03   0.834 0.404156
## ResidualSugar        1.095e-04  1.506e-04   0.727 0.467325
## Chlorides           -3.719e-02  1.604e-02  -2.319 0.020399 *
## FreeSulfurDioxide    9.333e-05  3.422e-05   2.727 0.006388 **
## TotalSulfurDioxide   8.333e-05  2.204e-05   3.781 0.000156 ***
## Density             -2.598e-01  1.918e-01  -1.355 0.175449
## pH                  -1.198e-02  7.540e-03  -1.589 0.112136
## Sulphates           -1.373e-02  5.465e-03  -2.512 0.012019 *
## Alcohol              3.386e-03  1.374e-03   2.465 0.013704 *
## LabelAppeal-1        2.353e-01  3.799e-02   6.195 5.84e-10 ***
## LabelAppeal0         4.256e-01  3.705e-02  11.487  < 2e-16 ***
## LabelAppeal1         5.578e-01  3.769e-02  14.799  < 2e-16 ***
## LabelAppeal2         6.970e-01  4.244e-02  16.424  < 2e-16 ***
## AcidIndex           -7.960e-02  4.576e-03 -17.395  < 2e-16 ***
## STARS2               3.191e-01  1.436e-02  22.226  < 2e-16 ***
## STARS3               4.384e-01  1.562e-02  28.060  < 2e-16 ***
## STARS4               5.585e-01  2.167e-02  25.770  < 2e-16 ***
## STARSunranked       -7.665e-01  1.954e-02 -39.226  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(40803.08) family taken to be 1)
##
##     Null deviance: 22860  on 12794  degrees of freedom
## Residual deviance: 13632  on 12774  degrees of freedom
## AIC: 45619
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  40803
##           Std. Err.:  34305
## Warning while fitting theta: iteration limit reached
##
##  2 x log-likelihood:  -45574.65
```

Negative Binomial Model with only chemical variables

```
Model_N2<-glm.nb(TARGET~.-FixedAcidity -VolatileAcidity -CitricAcid -ResidualSugar -Chlorides -FreeSulfu
```

```
## Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
## control$trace > : iteration limit reached
## Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
## control$trace > : iteration limit reached
```

```
summary(Model_N)
```

```
##
## Call:
## glm.nb(formula = TARGET ~ ., data = train_mice, init.theta = 40803.08215,
##     link = log)
##
## Coefficients:
##                       Estimate Std. Error z value Pr(>|z|)
## (Intercept)          1.464e+00  2.057e-01   7.114 1.13e-12 ***
## FixedAcidity         3.662e-05  8.199e-04   0.045 0.964371
## VolatileAcidity     -3.039e-02  6.529e-03  -4.655 3.25e-06 ***
## CitricAcid           4.921e-03  5.899e-03   0.834 0.404156
## ResidualSugar        1.095e-04  1.506e-04   0.727 0.467325
## Chlorides           -3.719e-02  1.604e-02  -2.319 0.020399 *
## FreeSulfurDioxide    9.333e-05  3.422e-05   2.727 0.006388 **
## TotalSulfurDioxide   8.333e-05  2.204e-05   3.781 0.000156 ***
## Density             -2.598e-01  1.918e-01  -1.355 0.175449
## pH                  -1.198e-02  7.540e-03  -1.589 0.112136
## Sulphates           -1.373e-02  5.465e-03  -2.512 0.012019 *
## Alcohol              3.386e-03  1.374e-03   2.465 0.013704 *
## LabelAppeal-1        2.353e-01  3.799e-02   6.195 5.84e-10 ***
## LabelAppeal0         4.256e-01  3.705e-02  11.487  < 2e-16 ***
## LabelAppeal1         5.578e-01  3.769e-02  14.799  < 2e-16 ***
## LabelAppeal2         6.970e-01  4.244e-02  16.424  < 2e-16 ***
## AcidIndex           -7.960e-02  4.576e-03 -17.395  < 2e-16 ***
## STARS2               3.191e-01  1.436e-02  22.226  < 2e-16 ***
## STARS3               4.384e-01  1.562e-02  28.060  < 2e-16 ***
## STARS4               5.585e-01  2.167e-02  25.770  < 2e-16 ***
## STARSunranked       -7.665e-01  1.954e-02 -39.226  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(40803.08) family taken to be 1)
##
##     Null deviance: 22860  on 12794  degrees of freedom
## Residual deviance: 13632  on 12774  degrees of freedom
## AIC: 45619
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  40803
##           Std. Err.:  34305
## Warning while fitting theta: iteration limit reached
##
```

```
##  2 x log-likelihood:  -45574.65
```

Predictions

```
train_pred1 <- predict(Model_p1, newdata = train_mice)
test_pred1<- predict(Model_p1, newdata = test_imp)
```

```
train_pred2 <- predict(Model_p2, newdata = train_mice)

test_pred2 <- predict(Model_p2, newdata = test_imp)
```

```
train_pred3 <- predict(Model_M, newdata = train_mice)

test_pred3 <- predict(Model_M, newdata = test_imp)
```

```
train_pred4 <- predict(Model_M2, newdata = train_mice)

test_pred4 <- predict(Model_M2, newdata = test_imp)
```

```
train_pred5 <- predict(Model_N, newdata = train_mice)

test_pred5 <- round(pmax(predict(Model_N, newdata = test_imp),0))
```

```
train_pred6 <- predict(Model_N2, newdata = train_mice)

test_pred6 <- round(pmax(predict(Model_N2, newdata = test_imp),0))
```

Comparsion of Models

```
# RMSE & MAE
rmse_p1 <- rmse(train_mice$TARGET, train_pred1)
mae_p1  <- mae(train_mice$TARGET, train_pred1)

rmse_p2 <- rmse(train_mice$TARGET, train_pred2)
mae_p2  <- mae(train_mice$TARGET, train_pred2)

rmse_m3 <- rmse(train_mice$TARGET, train_pred3)
mae_m3  <- mae(train_mice$TARGET, train_pred3)

rmse_m4 <- rmse(train_mice$TARGET, train_pred4)
mae_m4  <- mae(train_mice$TARGET, train_pred4)

rmse_m5 <- rmse(train_mice$TARGET, train_pred5)
mae_m5  <- mae(train_mice$TARGET, train_pred5)

rmse_m6 <- rmse(train_mice$TARGET, train_pred6)
mae_m6  <- mae(train_mice$TARGET, train_pred6)
```

```
# AIC
aic_p1 <- AIC(Model_p1)
aic_p2 <- AIC(Model_p2)
aic_m3 <- AIC(Model_M)
aic_m4 <- AIC(Model_M2)
aic_m5 <- AIC(Model_N)
aic_m6 <- AIC(Model_N2)
```

```r
# R-squared
# For linear regression
rsq_m3 <- summary(Model_M)$r.squared
rsq_m4 <- summary(Model_M2)$r.squared

# For Poisson regression (pseudo R²)
rsq_p1 <- 1 - (Model_p1$deviance / Model_p1$null.deviance)
rsq_p2 <- 1 - (Model_p2$deviance / Model_p2$null.deviance)

#Negative Binomial
rsq_m5 <- 1 - (Model_N$deviance / Model_N$null.deviance)
rsq_m6 <- 1 - (Model_N2$deviance / Model_N2$null.deviance)
```

```r
model_metrics <- data.frame(
  Model = c("Poisson Regression",
            "Poisson Regression Model 2",
            "Linear Regression",
            "Linear Regression Model 2",
            "Negative Binomial",
            "Negative Binomial Model 2"),
  RMSE  = c(rmse_p1, rmse_p2, rmse_m3, rmse_m4,rmse_m5,rmse_m6 ),
  MAE   = c(mae_p1, mae_p2, mae_m3, mae_m4,mae_m5, mae_m6),
  AIC   = c(aic_p1, aic_p2, aic_m3, aic_m4, aic_m5, aic_m6),
  R_squared = c(rsq_p1, rsq_p2, rsq_m3, rsq_m4, rsq_m5, rsq_m6)
)


kable(model_metrics, caption = "Comparison of Regression Models with RMSE, MAE, AIC, and R-squared") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"), full_width = F)
```

Table 1: Comparison of Regression Models with RMSE, MAE, AIC, and R-squared

Model	RMSE	MAE	AIC	R_squared
Poisson Regression	2.589648	2.221370	45616.23	0.4036886
Poisson Regression Model 2	2.590599	2.222448	45661.74	0.4009105
Linear Regression	1.304377	1.023543	43154.56	0.5414764
Linear Regression Model 2	1.310227	1.028302	43251.07	0.5373542
Negative Binomial	2.589648	2.221370	45618.65	0.4036815
Negative Binomial Model 2	2.590599	2.222448	45664.16	0.4009032

Based on the model comparison metrics, the linear regression model with all variables performed the best. This model achieved the lowest RMSE, MAE, and AIC, and the highest R-squared compared to the other models which were the Poisson, negative binomial and reduced-variable models. The linear regression may have done the best because the target variable may have had approximately linear relationships with the predictor variables causing the linear regression to capture the patterns. The full linear regression model balances simplicity and explanatory power, capturing most of the variability in wine cases purchased, which explains why it outperformed the other models in all metrics.

```r
test_imp$TARGET<-round(pmax(test_pred3))
```

```r
write.csv(test_imp, "Wine_Prediction.csv", row.names = FALSE)
```